Pranav R

Dual Degree in Engineering Design, **Indian Institute of Technology Madras**
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Engineering Design student focused on robotics, controls, embedded systems and autonomous systems, with experience developing multi-robot platforms, autonomous vehicles, motor-control systems and simulation frameworks from modelling through hardware validation.

Education

Degree	Institute	Year	CPI / %
Dual Degree in Engineering Design	Indian Institute of Technology Madras	2029 (expected)	**9.32**
Class XII	Emerald International School	2024	**95.8%**
Class X	Vidyaniketan Public School	2022	**98.9%**

Technical Skills

Robotics & Controls: ROS 2, Gazebo, MuJoCo, ArduPilot, PX4, Path Planning (A*, RRT*), Obstacle Avoidance, PID, Model Predictive Control, RL-based Control, Extended Kalman Filters (EKF), Sensor Fusion, Perception

Embedded Systems & Hardware: STM32, Arduino, ESP32, PCB Design, CAN Bus, Motor Control, Hardware-in-the-Loop (HIL)

Programming, ML & Computer Vision: Python, C/C++, PyTorch, TensorFlow, CNNs, OpenCV, Reinforcement Learning (TD3)

Simulation, CAD & Analysis: MATLAB/Simulink, Fusion 360, Ansys Finite Element Analysis (FEA)

Tools & Platforms: Linux, Git, pytest, OptiTrack Motion Capture

Professional Experience

Software Intern – Multi-Robot Simulation & Task Allocation *Shunyaanantam*
[Feb 2026] – Present

- Built a reusable multi-drone simulation package **(ROS 2, Gazebo, ArduPilot SITL)** with YAML-configured N-vehicle spawning, a multi-platform vehicle registry, formation helpers and a GUI launcher, used to validate multi-agent consensus and formation control.
- Designed an N-drone decentralized task-allocation and delivery simulator **(CBBA + APF collision avoidance)** in MATLAB/Simulink, achieving **100% conflict-free mission completion** with dynamic re-tasking and battery-constrained return-to-base.
- Built a modular benchmark for centralized vs. decentralized multi-drone path planning **(A*, RRT*, BiRRT*, PRM)** across environments, fleet sizes and communication topologies, with independent safety evaluation.

Research Projects

Multi-UAV Formation Control *[Aug 2026] – Present*
Control Engineering Lab (Prof. Bharath Bhikkaji) & Distributed Intelligence and Robotics Lab (Prof. Anuj Tiwari), IIT Madras

- Investigating hardware realization of a published multi-UAV beamforming framework using **Crazyflie nano-UAVs** with closed-loop motion-capture feedback.
- Extended the formation-control formulation to **nonlinear 6-DOF quadrotor dynamics**, incorporating roll, pitch, thrust, velocity and actuator constraints.
- Formulating constrained trajectory execution to minimize both **tracking error and beam-pointing error**, with simulation-to-hardware validation using OptiTrack.

Robotics Projects

Project CLIMB – Autonomous Pipe Inspection Robot *iBOT, IIT Madras*
Project Lead *2026 – Present*

- Pitched and founded CLIMB, leading a 7-member team across perception and mechatronics, building robotic inspection coupled with **digital twins for predictive maintenance** for oil & gas **(ONGC)** and maritime infrastructure.
- Architected the magnetic adhesion and mobility system to maintain stable contact with ferromagnetic pipelines across varying orientations while ensuring reliable locomotion.
- Validated magnetic adhesion forces using **ANSYS Maxwell** and developed a **Gazebo physics** model for adhesion, traction and articulated locomotion.
- Building the perception and inspection pipeline for visual corrosion detection **(94% accuracy)** and non-destructive thickness measurement using ROS 2 and computer vision.

Sensorless Precision Insertion for a SCARA Manipulator *Analog Devices India – Anveshan*
Team of 4 *2026 – Present*

- Building a sensorless precision-insertion capability for a SCARA arm, inferring contact and insertion state purely from **motor current signatures**, without dedicated force/torque or position sensors.
- Investigating **DSP-based sensorless estimation** of actuator state and contact conditions from motor-driver signals, targeting sim-to-real validation on ADI MAX32690/TMC2209 hardware.

Team Abhiyaan – Project Vikram, Autonomous Ground Vehicle (IGVC, USA) *2025 – Present*
Mechatronics Engineer

- Established **CAN-based** communication between the vehicle's central controller and motor-control hardware for reliable command and telemetry exchange.
- Implemented and tuned cascaded **PID** velocity control, achieving stable motion with **sub-5% overshoot**.
- Engineered kinematic path-planning and parallel-parking algorithms, validated in simulation with **ROS 2, Gazebo and Nav2**; designed an event-aware pedestrian-alert horn system for an autonomous golf-cart platform.

GPS-Denied Nano Drones *Inter IIT Tech Meet 14.0 – Bronze Medal*
Drona Aviation problem statement *[Dec 2025]*

- Implemented **EKF**-based sensor fusion for stable autonomous flight of nano drones in GPS-denied indoor environments.
- Tuned **cascaded flight-control** loops and optimized the Magis firmware, achieving **centimetre-level hover precision** without external localization.

ASCEND – Autonomous Aerial Surveyor *ISRO Robotics Challenge – Top 20 / 600*
[Jan - Jul 2026]

- Designed and integrated an autonomous charging and docking station for drones, enabling sustained, long-duration operations.
- Developed an automated feature localization pipeline using **DINOv2 ,PyTorch and ROS2** , implementing dense cosine similarity matching and attention-map weighting to detect physical landmarks across aerial flight images.
- Increased pipeline efficiency and accuracy by filtering blurry frames via **Laplacian variance, clustering stop-group detections**, and enforcing majority voting to output structured YAML coordinate maps.

RoboCup Small Size League (SSL) – Team Abhiyaan *IIT Madras*
[Jun 2025]

- Engineered the communication topology and centralized networking framework for real-time, low-latency coordination across a team of 6 autonomous robots.
- Designed the electrical subsystem and **custom PCB** to power and control a high-impact kinetic kicking mechanism.

Controls & Embedded Systems

RL-Based Active Disturbance Rejection *Caterpillar Tech Challenge – Winner*

- Implemented RL-based feedback (**TD3**) for adaptive gain tuning of PI controllers, reducing overshoot/undershoot **from 40% to under 5%** and cutting settling time by over **80%** via a custom replay buffer and oracle-bootstrapped training.

Project Neo-G1 – Smart Neonatal Incubator *Provisional Patent Filed (IDF 3766)*
Sahaay, IIT Madras – supervised by Prof. Tiju Thomas *Oct 2025 – Apr 2026*

- Designed the **electrical and safety architecture** of a low-cost, modular incubator for resource-constrained hospitals, including the sensor suite, PCB and closed-loop temperature and humidity control.
- Modeled the thermal and control system in MATLAB/Simulink to validate the controller.

Other Projects

- **Project Agilit-e** (Course Project, *Complete Patent Filed*) – Led modeling and FEA-based structural analysis of the steering subsystem of a narrow three-wheeled urban vehicle's suspension and mobility system.
- **Project skincAIr** (Sahaay) – Built a CNN skin-condition classifier (**92%** accuracy) with an LLM-based structured reporting layer, in Python and TensorFlow.

Leadership & Positions of Responsibility

- **Associate Manager & Core Team**, E-Cell IIT Madras (2024–25) – analytics; ran the Unconference event.
- **Deputy Coordinator**, iBOT and Sahaay, IIT Madras (2024–25); **Saathi Student Mentor** to 4 freshmen (2025–Present).

Achievements

- **Caterpillar Tech Challenge 2026** – Winner · **Inter IIT Tech Meet 14.0** – Bronze (23 IITs) · **ISRO Robotics Challenge 2026** – Top 20 of 600
- **Hult Prize India Nationals 2026** – Top 80 of 1,500+
- JEE Advanced AIR 2506 (top 1%) · KCET Rank 61 (of 200,000+)
- Top 100 Young Science Talents in India (SPOT100 GenII, Vikram Sarabhai Space Foundation);